SWAPNIL MITRA

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Software engineer specialising in cloud integration, backend systems, and AI-enabled tooling. I build dependable platforms that connect complex systems and take them from architecture through to production.

TECHNICAL SKILLS

Languages
C#, TypeScript, JavaScript, Python, Rust, SQL, Solidity

Frameworks
.NET 8, ASP.NET Core, React, Next.js, Flutter, Express, Fluent UI, Tailwind CSS

AI & Data
LangChain, AWS Bedrock, RAG pipelines, pgvector, SQL Server, PostgreSQL, MongoDB, Prisma, Drizzle

Azure
Functions, API Management, Logic Apps, App Service, Entra ID, Key Vault, Application Insights, Monitor

DevOps & Messaging
GitHub Actions, Azure DevOps, Docker, RabbitMQ, Prometheus, Grafana

Platforms & Tools
AWS (S3, Transcribe, Polly), Microsoft Graph, SharePoint, Hardhat, IPFS, Git

PROFESSIONAL EXPERIENCE

Adrosonic | **Research and Innovation Engineer** · *Jul 2023 – Present*

Insurance Broker Management Platform

- Led a team of 4 on a broker management platform (.NET, SQL Server, MongoDB on Azure), used daily by insurance brokers across Australia and New Zealand to quote, bind, amend, and cancel policies.
- Delivered Microsoft single sign-on and transactional policy updates, so no policy change is ever left half-processed.
- Architected the asynchronous integration layer connecting the platform to insurers on Azure API Management, Logic Apps, and Functions, cutting integration time from ∼6 weeks to ∼2 with 99.9% successful delivery.
- Added end-to-end tracking and alerting with Application Insights and Azure Monitor; wrote the handover docs.

Policy Administration Integration

- Led delivery of a single API on Azure Functions giving external callers one interface to several policy administration systems, routing each request to the right system, merging results, and handling slow writes asynchronously.
- Built on stateless per-system adapters behind one shared contract, so new consumers onboard without API changes.

Release Engineering

- Led the team that built a shared, versioned CI/CD framework on GitHub Actions that any repository adopts with minimal config, standardising build, test, and release for .NET, Azure Functions, NuGet, and Logic Apps.
- Added release safeguards: pull-request quality gates linked to Azure DevOps work items, approval-gated Staging and Prod releases with fast slot-based rollback, OIDC-based Azure access, and Key Vault-only secrets.

Email Filing & Document Management

- Rebuilt an email filing and SharePoint document pipeline on .NET and RabbitMQ publish/subscribe messaging with duplicate protection, dead-letter queues, and Prometheus monitoring, cutting message latency ∼40%.
- Replaced a legacy Outlook add-in with a React and Fluent UI web add-in with Microsoft sign-in, Graph integration, and offline caching, rolled out to 10,000+ users.

PROJECTS

- **MemVault** · *Aug 2026* Local-first memory engine for AI agents in Rust with a tamper-evident record of everything stored and retrieved, provable deletion, hybrid semantic and keyword search, and MCP and Python bindings.
- **Groundwork** · *2024 – 2026* AI outreach workspace that reads a job posting and drafts evidence-backed emails with an LLM over hybrid search, gated by human approval and an audit trail (Next.js, Postgres + pgvector).
- **City Vault** · *2025 – 2026* Multi-tenant records management platform with six permission roles, full document lifecycle, and an append-only audit trail backed by IPFS storage (Next.js 15, Prisma, Postgres).
- **AGRISETU** · *Feb – Apr 2026* Voice-first AI marketplace (Flutter, Next.js, Express) where farmers order supplies together by voice in 11 languages, built on AWS Transcribe, Polly, a Bedrock LLM, and a RAG knowledge base.

EDUCATION

Indian Institute of Technology, Kharagpur | **B.Tech, Biotechnology and Biochemical Engineering** · *May 2023*

EXTRA-CURRICULARS

- Associate Manager, E-Cell IIT KGP: organised Global Entrepreneurship Summit 2021, LSM, EAD, Empresario 2020.
- National Service Scheme Volunteer, IIT KGP: Silver Medal for collaborative performance (2019).
- Club Service Director, Rotaract Club of Calcutta North East.